LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Unit Test: Combustion and Flame

Chapter / Topic: Combustion and Flame (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 45 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Which of these is a combustible substance? **[1]**

(a) Glass (b) Stone
(c) Iron nail (d) Wood

Q2. The hottest part of a candle flame is the: **[1]**

(a) outermost non-luminous zone (b) innermost dark zone
(c) middle luminous zone (d) wick

Q3. Which is the best fire extinguisher for a fire in electrical equipment? **[1]**

(a) Water (b) Carbon dioxide
(c) Kerosene (d) Petrol

Q4. The unit of calorific value is: **[1]**

(a) kg/kJ (b) kJ/kg
(c) °C (d) joule only

Q5. Which of these fuels has the highest calorific value? **[1]**

(a) Wood (b) Coal
(c) Hydrogen (d) LPG

Q6. The burning of a candle involves: **[1]**

(a) only a physical change
(b) only a chemical change
(c) both physical and chemical changes
(d) no change

SECTION B

8 marks

Q7. What is ignition temperature? Why does wet wood not catch fire easily? **[2]**

Q8. Why should water not be poured on a fire caused by oil? **[2]**

Q9. What is spontaneous combustion? Give one example. **[2]**

Q10. Name the poisonous gas formed when fuels burn incompletely. Why is it dangerous? **[2]**

SECTION C

6 marks

Q11. 3 kg of a fuel produces 150,000 kJ of heat on complete burning. Find its calorific value. LPG has a calorific value of about 55,000 kJ/kg. Which of the two is the better fuel by this measure? **[3]**

Q12. Name the three zones of a candle flame and describe what happens in each. **[3]**

SECTION D

5 marks

Q13. Answer the following: **[5]**

(i) What three things are needed for a fire to start and continue? **[2]**

(ii) How does a carbon dioxide fire extinguisher put out a fire? **[1]**

(iii) Give two harmful effects on the environment of burning wood, coal and leaves. **[2]**

~ All the best! ~